E4 Final Prototype Memo

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Background

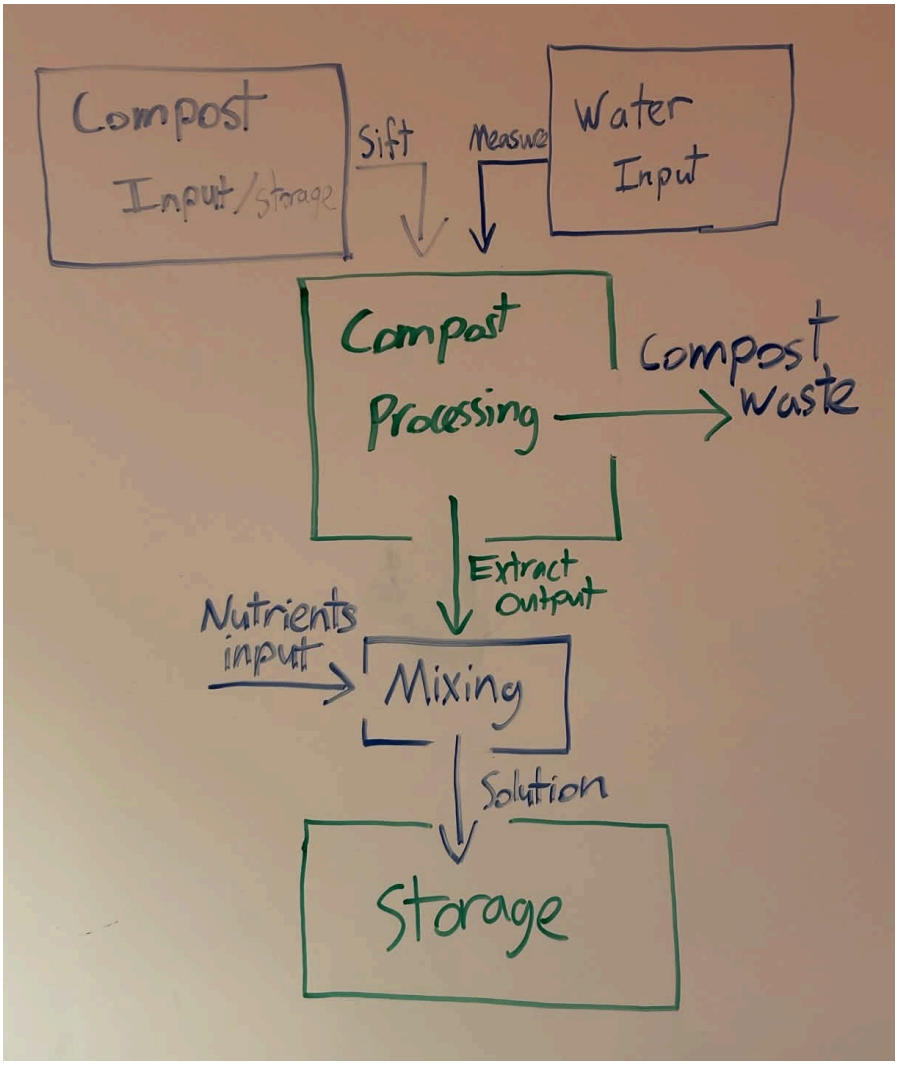


Figure 1: Abstraction diagram of compost extraction process

Our client, Ty Tessiere, produces compost extract containing beneficial microorganisms that are injected into soil at agricultural and residential sites for his personal business, *Vermillion Compost Soil Solutions*. The entire compost extraction process is outlined in Figure 1. However, Ty's current extraction and injection process presents several challenges. Currently, Ty uses individual mesh bags, putting three scoops of compost in each bag. He sprays these three scoops with water once more, before switching them out for three more scoops and repeating the same process until he has sprayed 20 gallons of compost. After extraction, Ty mixes the compost extract with nutrients and water. Once he is finished with this process, he then injects the compost extract into the ground. The time-intensive nature of this process restricts the number of job sites Ty can visit per day (Teissere, Client Meeting Number 1).

Problem Statement

The client seeks a design that supports the preparation and underground injection of compost extract while ensuring minimal preparation time, minimal manual effort, effective drainage, consistent microorganism dispersion, and allowing them to perform multiple jobs in a single day without returning home.

Prototype Question & Description

Question: Does the height of our spray nozzle provide enough pressure to make a good-quality compost extract?



Figure 2: Photo of compost extractor

The prototype we are testing is a fully functional prototype involving all components of our system. This prototype involves a motorized auger powered by a DC motor. The auger is held in a mesh tube made from aluminum rings and zinc-plated bolts, lined with 400 micron mesh. The auger sleeve fits into a liquid collection tank made from polycarbonate. The liquid collection tank is attached to an aluminum motor housing. Above the motor housing sits the compost bucket that funnels compost into the back end of the auger. Finally, a PVC pipe scaffolding system holds a fan-tip spray nozzle above the auger.

Final Functional Design Justification: Objectives and Constraints

Objectives

- Minimize time spent in preparation at the site:

As an automated system, the prototype does meet this objective. The only non-automated part of the system is the setup and the compost entry for which the user is required to manually agitate the compost in the funnel.
A major component of the design is the motorized auger. This auger continuously pushes compost through the system, requiring no manual work. In the evaluation results section of this memo, evidence is provided that the motorized auger works.

Objective	**Metric: Quantitative - Time spent on setup and system maintenance time during a compost extraction run**	
Minimize Time Spent in Preparation at job site	Type of Scoring: Scale from 0-5	
	Description:	Score
	> 20 minutes	1
	15-20 minutes	2
	10-15 minutes	3
	5-10 minutes	4
	< 5 minutes	5

Table 1: Metrics for minimizing time spent in preparation at job site

- Maximize concentration of microorganisms in compost extract:

  A major part of our design is the half-moon auger sleeve. Because the top third of the auger sleeve is open, exposing the auger and compost, water sprayed from above can more easily contact the compost. Since our system uses a flat spray nozzle, it can evenly spray every part of the auger, guaranteeing that the compost being dispensed by the auger will always come in contact with water and will be sprayed. Along with this, the auger is rotating at a relatively slow rate, which will allow the compost to have more contact time with the water. In our evaluation results, evidence is presented that the compost extract is highly concentrated and even in concentration. The color is homogeneous throughout and very dark. Evaluating concentration of extract is done visually using the below color chart. Similar studies find color to be an appropriate indicator of compost stability and potency (Ashik et al.).

Objective	**Metric: Qualitative - Concentration of extract judged by its color**	
Maximize amount of microorganisms knocked off compost	Type of Scoring: Scale from 0-5	
	Description:	Score
		1
		2

		3
		4
		5

Table 2: Metrics for maximizing amount of microorganisms knocked off compost

- Maximize drainage

  The liquid casing going into the tank has a ramp angled down to the hole, maximizing drainage. In addition to the ramp, the liquid casing has sealant applied at all crevices, preventing leakage. This prevents any compost extract from being stuck at the bottom of the liquid casing. In comparison with our client's old system, where there was always a pool of excess compost extract, our current system does not have any stagnant extract, allowing for no opportunities for bad bacteria or mold growing and affecting the overall quality of the final product.

Objective	**Metric: Quantitative - Amount of extract left in liquid casing after spray (ml)**	
Maximize drainage	Type of Scoring: Scale from 0-5	
	Description:	Score
	>3ml	1
	2-3ml	2
	1-2ml	3
	0-1ml	4
	0ml	5

Table 3: Metric for maximizing drainage

- Maximize durability of the device

  Machined components of our system, like the waterjetted auger cage rings and polycarbonate liquid casing, ensure maximum durability of our system. It is important to consider that we maximized durability within the budget of $150 that we were provided. With a higher budget, the durability could have been further improved.

Objective	**Metric: Qualitative - Amount of wear experienced by prototype after 1 test**
Maximize	Type of Scoring: Scale from 0-5

durability of device	Description:	Score
	Prototype failure: A portion of the prototype breaking into unusable condition	1
	Major wear: Large gashes or cracks leading to almost unusable conditions	2
	Medium wear: Noticeable gashes or cracks but still in easily usable condition	3
	Minimal wear: Small scratches or nicks that have no apparent effect on prototype function	4
	No visible wear observed	5

Table 4: Metric for maximizing durability of device

- Minimize weight of device

  By making our system physically smaller than the tools Ty previously used, we were able to create a lighter system for him, making it much easier for him to load and handle. Along with this, we used very light materials like polycarbonate and PLS for the majority of the project to make sure our system didn't go over the truck's weight limit. For heavier materials, we only used them when we needed them for stronger, more reinforced parts of the design, but those ended up being used much less, making our system strong but also lightweight.
  In the future we could further minimize weight by using hollow bolts instead of solid ones and by substituting our aluminum casing for a lighter material.

Objective	**Metric: Quantitative - Mass in pounds**	
Minimize Weight of Device	Type of Scoring: Scale from 0-5	
	Description:	Score
	>140lbs	1
	100-140lbs	2
	60-100lbs	3
	20-60lbs	4
	<20lbs	5

Table 5: Metric for minimizing weight of device

Constraints

After the initial client meeting and further follow-up questions with the client, a revised list of constraints was created, which is as follows:

- Must be smaller than dimensions: Length: 73.5 in Width: 53 in (Galaxy Toyota).
    At the end of our first client meeting, Ty shared the 1996 4-cylinder Tacoma he uses for his jobs. It has limited bed space to store whatever solution we create. During the design phase, we had to consider solutions that were portable and small enough to fit within our bed. Our final prototype assembly is 27 inches long and 14 inches wide, making it suitable to fit within his truck bed. The final product can also be disassembled to better fit into the truck bed.
- Must not weigh more than ~835 lbs
    In addition to a sizing constraint, Ty's truck is not able to handle heavy loads above the weight of 100 gallons of water. He shares his experiences of having trouble with braking and almost crashing when his truck was carrying 100 gallons of water plus the weight of several other devices. He requested that whatever device we create to not exceed the weight of 100 gallons, and ideally would not need to store any liquid at all. Our device does not need to store any water and furthermore weighs in at 16.4 lbs, well below the demanded constraint (Galaxy Toyota) .
- Must cost less than $150
    As the project was funded by Harvey Mudd College's Engineering Department, we were given the constraint of keeping the project below $150. However, we were able to scavenge resources throughout the department, including an abundance of PLA for prototyping. The final project spending ended at $148.31, indicating that this was the most challenging constraint the project faced. Although the bill of materials netted well over $150 (due to extensive scavenging), the constraint only applied to our project budget, indicating a success in meeting this constraint.
- Must not leave compost extract stagnant in the tank for more than 3 hours maximum
    After each job, Ty explains that he must thoroughly clean all of his systems to ensure that all stray compost extract is removed. Compost extract, when left stagnant for too long, becomes anaerobic and develops large quantities of poor bacteria that could be damaging to plants (Liu et al.) . As our prototype continuously makes compost, it will not be necessary for the compost to sit for a long amount of time; Ty can begin to inject his extract simultaneously as the extract is created. Additionally, our prototype uses plenty of sealant to avoid extract falling into difficult-to-reach crevices, and is easily detachable, making it possible for Ty to clean the system himself after a job.
- Must maintain pH within the range of 5.5- 7.0 within the device
    The microorganisms within the compost extract are sensitive to fluctuations in pH, and a pH beyond the indicated range could result in the death of many of the important microbes. Our system works around this constraint by only allowing nutrients to be input while Ty is injecting the extract. As nutrients are very acidic, this workaround allows the nutrients to be mixed with the extract only once it has left the tank, so that it does not rest in a high pH solution. Additionally, the rate at which nutrients are input is regulated by a tuned venturi system that allows the nutrients to be diluted prior to mixing.

Final Functional Design Justification: Material Choices

Auger Cage

- Aluminum plates
    Aluminum was utilized because the auger cage needed to be very rigid to sustain constant wear and laterally support the steel auger. Previous prototypes utilized PLA plates; these were flexible and did not provide sufficient support for a permanent model. Thus, we chose aluminum plates to maximize the durability of the device.
- PLA standoffs

The standoffs between each plate only carry uniaxial load and are primarily to ensure the plates remain equally spaced across. As this uniaxial load is very limited (simply the pressure exerted by the nuts and bolts fastening), we opted for printing the standoffs out of PLA to save on manufacturing time, stay under $150, and minimize the weight of the device.

- ⅝” Carbon Steel Bolts

  Long bolts were utilized to connect each plate. These bolts needed to be made out of steel as they carry most of the load of the compost and auger, netting to roughly 8lbs during extraction. This was to ensure that the durability of the device is maximized.

- 400 Micron mesh + stiff metal mesh

  400 micron mesh was the sizing Ty had used previously with his manual extraction methods (Teissere, Client Meeting Number 1) . In early prototyping phases, we identified that this size mesh worked phenomenally for producing an extract of a high concentration. Testing also revealed that the mesh needed support from a larger mesh size, hence the usage of a stiff metal mesh surrounding the fine mesh. We chose the final combination of mesh to maximize even dispersion of microorganisms in the compost extract.

- Steel Auger

  The client, Ty, requested that steel, ideally stainless steel, be used for important parts of the system (Teissere, Client Meeting Number 2) , as it has a reputation for being durable and water-resistant. As the auger is central to our system's operation, we decided to invest ~$18 into purchasing a steel auger for the extraction process. This was partially because previous prototypes utilized a PLA auger and were prone to snapping.

Liquid Casing

- Polycarbonate Walls

  As the client requested that the prototype drains easily to avoid anaerobic bacteria, we found that it would be helpful for the casing to be primarily made of a transparent material. Additionally, as the auger and its sleeve must rest on whatever material the walls were made out of, the material must be durable and strong. Polycarbonate was chosen for the job as it is transparent, lightweight, extremely strong, can be cut on a CNC machine, and would produce a sleek finish for the final design (Hafad et al.) . Our design choice follows the two objectives: Maximize durability and minimize weight.

- J.B. Weld Epoxy

  Initially, we considered welding the edges of the cage together permanently using a solvent such as Weld-On for acrylic and plastics. However, due to budget constraints, we opted to simply fasten the casing with epoxy and angle brackets. A two-part general epoxy “J.B. Weld” was used due to its availability within the resources we were able to scavenge and its reputation for strength/durability.

- Aluminum Angle Brackets

  The epoxy was not sufficient for fastening the liquid casing together - mechanical fasteners were necessary to ensure the epoxy bonds would not rip apart under tension. Aluminum was opted for due to its lightweight and appropriate rigidity, given the size of the casing.

- Silicon sealant

  To avoid leaks, a silicone sealant was applied to all edges of the casing. Silicone was selected as it is safe for plants and would not contaminate any of the resultant compost extract.

- Duct Tape

  In order to place the cage into the polycarbonate liquid casing, we had designed it to be completely open, allowing the cage to sit above the top of the casing. However, since we need to ensure all the water passing through the system is passed through the compost first, we decided to duct tape the sides of the liquid casing to prevent any spray of water from the hose from accumulating in the extractor

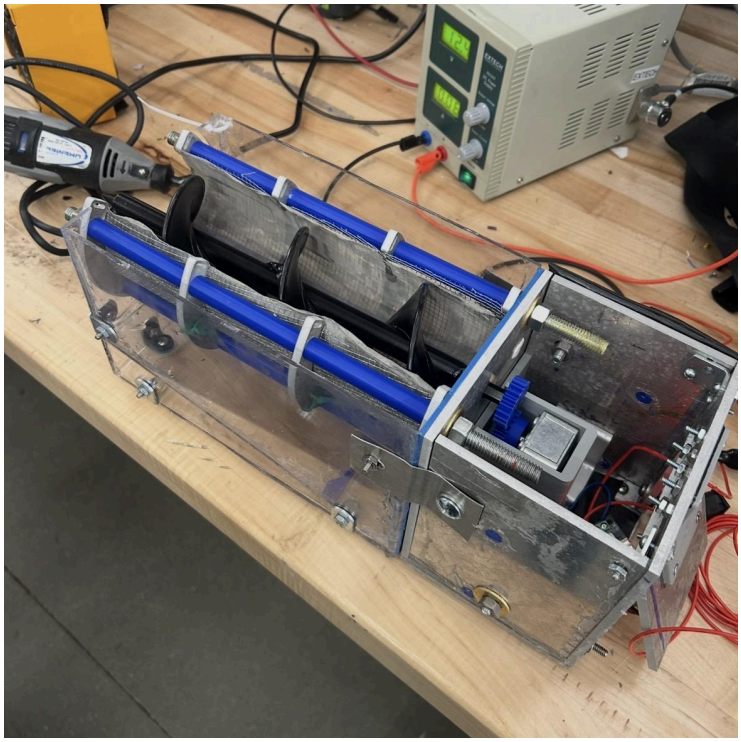


Figure 3: Photo of liquid casing and motor casing before compost was input

Motor Casing

- Duct Tape

  While making the walls of the motor casing, we ran out of aluminum and time to create a lid for the motor box, which left the entire top of the compost exposed. In order to prevent compost (which sits above it) and water (which sprays down next to it) from getting to the electronics, the quickest and most reliable solution we could implement for testing was duct tape.
- Aluminum plates

  Since it was originally designed to bear the load of all of the compost by connecting to some additional tank or vessel as a compost in, the motor casing material was chosen to be aluminum instead of polycarbonate. Overall, the process of cutting and designing the pieces was similar to that of the polycarbonate walls, so we opted for the aluminum for its strength.
- PLA Motor mounts

  The motor had quite an irregular shape, given its cylindrical lower body and rectangular upper gearbox. To support the motor in place as it spun and transferred rotational energy to the auger gear, we modeled a custom motor housing. As many of the parts were very irregular and the housing did not particularly need the strength of metal, we opted to print the mount out of PLA.
- Various Angle bracket hardware

  Since we had aluminum plates as the walls of the motor box and we didn't have the capabilities to weld, we needed a secure way to attach the door and frame together. So, using scavenged angle bracket parts, we drilled various holes into the box and secured each with a nut and bolt.
- Silicon Sealant

  The motor box houses our electronics and motors, so it was important to make sure the system was as watertight as possible. Thus, we sealed all of the holes and edge gaps in the box to prevent water from getting in.

Compost in

- Home Depot bucket

  It was our client's request to create a solution that could process up to 5 gallons (one bucket's worth) of compost per cycle. Instead of creating a vessel that could contain that much compost and integrate with our system, we decided to alter one of our current buckets since it was already the proper size limit, and it was much easier (and cheaper) than making a tank from scratch.
- PLA Funnel
  - We liked the idea of generating a unique shape in SolidWorks to better fit our needs rather than purchasing and acquiring one with a very specific geometry, so we decided to go with 3D printing for the funnel. Since the funnel sits into the bucket and compost is passing through the opening rather than generating a load, we figured PLA would work well enough for our purposes.
- Wooden Blocks

  In order to raise the top-heavy bucket above the extractor without it tipping over (compost included), we chose to screw wooden blocks under the bottom for a stable configuration. The wooden blocks themselves were scavenged, which helped us stay within the $150 constraint.



Figure 4: Bucket and funnel system for compost entry system

Water in

- PVC Pipes

  We needed a lightweight, easy-to-make, and adjustable support for the water-in system so we decided to attach various PVC pipes and pipe couplers from the design studio to the compost-in system. The result was a simple arm/frame spanning across the extractor that is easy to assemble and disassemble.
- Various Hose attachments

  The various hose-to-pipe fittings we needed to assemble the water-in system were very specific and would have been too expensive to procure. So we found the parts needed by scavenging in the engineering stockroom drawers.

Supports and table

- Plywood

  The table that goes underneath the extractor to hold it in place is mostly supported by the ring of the tank. Therefore we opted for a thinner material, like plywood, that was easier to sketch on and cut for manufacturing purposes.
- PLA Standoffs

  The standoffs on the table act as alignment tools and experience little to no load overall. Thus we reused old (previously incorrectly sized) PLA standoffs for efficiency's sake.

Prototype Functionality

The following list of functions that were previously outlined in the checkpoint prototype memo will be sorted into primary and secondary functions. For each function, the final prototype will be evaluated in terms of its ability to accomplish that function and any strengths or weaknesses noted during the testing process.

Primary:

- Saturate compost with water
  - By having a specialized fan nozzle that is connected to a female-to-female hose coupler, the client can connect a hose to the connector that allows water to be transferred to incoming compost that is cycled through by the metal auger. This compost will be soaked by the water being sprayed by the nozzle and therefore saturated. This function is considered to be functional in our final prototype.
- Knock microorganisms off the compost
  - The fan nozzle effectively redirects water from a water source (i.e., hose) where outgoing water has sufficient water pressure to hit the compost extract and knock microorganisms into the filtered liquid. The amount of microorganisms in the filtered liquid or compost extract is judged qualitatively by the darkness of the liquid. This function is considered to be functional in our final prototype.
- Separate compost chunks from liquid extract
  - A fine 40 mesh/400 micron mesh successfully filters compost extract liquid from saturated compost while the auger pushes compost out the end of the auger sleeve. The saturated compost can be collected as waste in a container separated from the container containing compost extract. There will be no solid compost in the container intended to store compost extract. Therefore, this function is considered functional in the final prototype.
- Provide compost extract to the injector
  - Once the microorganisms are knocked off and filtered through the fine mesh, a ramp directs incoming compost extract into a small hole that would go into any container that would store compost extract. This container can either be the large 100-gallon tank in which the client creates compost extract or a different bucket that can be emptied into the 100-gallon container. Either way, compost extract is able to be stored and eventually supplied to the injector. This function is considered functional in the final prototype.

Secondary:

- Regulate water volume
  - No set volume or timer delivers an exact ratio of compost to water. However, the motor speed of 18 revolutions per minute provides a sufficient ratio of incoming water to incoming compost. This function is considered to be slightly successful as judged by the quality of compost extract derived from testing.
- Regulate compost volume

    - Compost can only be stored in the five-gallon tank in the compost-in subassembly. Although there is a 3-D printed funnel, the final prototype does not involve an automated method of regulating the amount of compost supplied to the rotating auger. The client would need to physically tap the five-gallon bucket in order for compost to flow through into the auger. The rotation of the auger at 18 revolutions per minute still provides good-quality compost extract. Therefore, future prototyping and modifications will be made with this function in mind.
- Incorporate compost, seaweed, and fish-based nutrients
    - A 3-D printed Venturi injector successfully draws liquid from an external source (that would contain the nutrients) using a pressure differential. Prototyping demonstrated that the venturi injector did not break due to the high pressure. Therefore, the client is able to include different types of nutrients in the final compost extract mix. This function is considered to be functional in the final prototype.
- Agitate compost extract
    - Additional research and prototyping informed us that this function was not necessary and very difficult to replicate in the final prototype. The group believed that an agitation system could not be included within the scope of the budget for the project.

The evaluation process of the final prototype informed us of many strengths and weaknesses of our project. Compost extract was able to be obtained in a semi-automated way, which was the primary goal. With a 12V voltage source and a water source, we believe that the client can use the final prototype as is, closely following a manual that describes how to use the device. Future testing and modifications will refine the prototype and cause it to be more efficient, clean, and simple.

Evaluation Plan

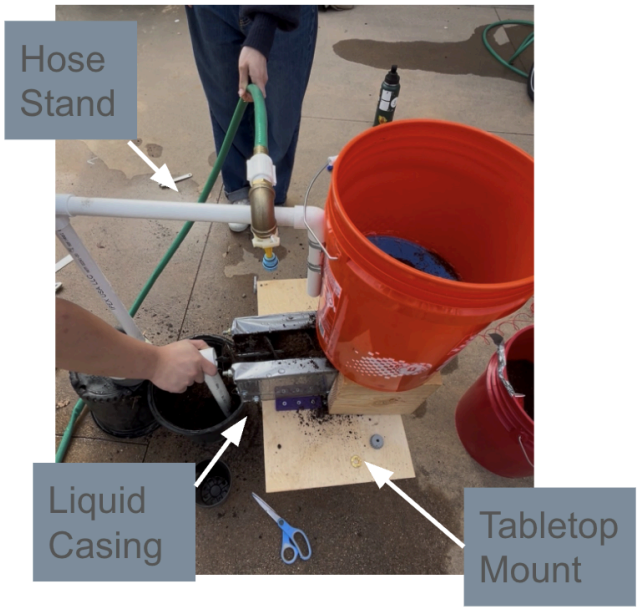


Figure 5: Labeled compost extractor

1. Place the tabletop mount on top of a 5-gallon bucket so that the interior standoffs brush against the interior edges of the bucket.
2. Ensure the motor is properly set into its mount, with the positive lead (indicated by the red dot) connected with a jumper wire to the “motor +” port of the motor regulator. Repeat for the negative terminal, and ensure the motor housing is properly sealed against water.
3. Place the auger casing and the auger into the liquid casing such that the bolts of the auger casing align with the holes in the motor housing, and fasten the bolts of the auger casing.
4. Place the motor housing and liquid casing on top of the tabletop mount such that the drainage hole from the liquid casing is placed above the hole in the tabletop mount.
5. Connect the motor to the power supply using alligator clips to connect the black marked wire to the ground of the power supply and to connect the fully red wire to the power output of the power supply.
6. Place the compost dispensing bucket above the auger such that the funnel out of the bottom of the bucket is directly above the end of the auger closest to the motor mount.
7. Screw a male garden hose into the female garden hose attachment to the spray nozzle, and fasten the spray nozzle to the PVC pipe hose stand using the plastic hose clamp.
8. Fill the bucket with 80 fl oz of compost (approximately 5 16 oz. solo cups)
9. Ensure the power supply is plugged in and is far from the direction of water spray. Turn on the power supply.
10. Turn on the motor in reverse by hitting the “II” switch on the back of the motor housing. The auger will begin spinning.
11. Wait 7 seconds, then turn on the spigot, allowing water to flow into the garden hose.
12. Continue spraying until all the compost has moved through the auger.
13. Turn off the water and analyze the compost extract quality by comparing it to the color chart in the metrics chart below. Record the compost extract score.
14. If the compost extract scores a 2 or higher, it is considered to be successful.
15. Repeat this process, starting from step 8, at least three times for each different height.



Figure 6: Resultant extract from Trial 3 of the second height. This corresponds to a score of 3

Objective	Metric: Qualitative - Concentration of extract judged by its color	
Maximize amount of microorganisms knocked off compost	Type of Scoring: Scale from 0-5	
	Description:	Score
		1
		2
		3
		4
		5

Table 6: Metric for maximizing amount of microorganisms knocked off compost

Evaluation Results

From the trials, we observed that when the spray nozzle was at a height of 11/12” above the auger, the compost color was at its darkest, achieving an average rounded score of 3, while the other two heights yielded an average score of 2. By positioning the spray nozzle closer to the auger, we can produce a higher-quality compost extract for our client. Providing the best possible extract for Ty’s business is vital, as a better quality extract will be more beneficial to the job sites where he is working. If the extract quality falls short, it could negatively impact his professional reputation and business.

During testing, we noticed that the compost would occasionally get stuck in the funnel and would stop depositing into the auger. This created a problem because whenever the auger stopped pushing material through, the sprayer kept running, which meant that there was just pure water being added to the bucket below, thus diluting the concentration of the final extract we were looking for. To account for this and ensure our data remained accurate, immediately when we noticed a clog, we stopped the test and re-did the trial for that certain height

Other inconsistencies were varying water pressure coming out of the hose and issues with spray aiming. Varying water pressure could cause changes in the compost extract quality. For example, if there was higher pressure at a point, it would knock off more microorganisms. Along with this, depending on the spray angle, there could have been more water bouncing off the auger casing or other areas, which would dilute the extract to an uncertain amount. To account for these uncertainties, we loosened the hose the exact same amount for each trial. Because

water pressure coming out of the hose is an environmental variable, we had to account for this by simply conducting multiple trials.

In the future, to ensure the compost flows into the auger more smoothly, we could design a funnel that eliminates sharp corners and leads straight into the intake. We noticed that the current sharp edges tend to catch the compost, causing it to stagnate rather than sliding down consistently. Along with this, the compost tended to clog the hole of the funnel, which forced us to manually unclog the hole, defeating the purpose of an automated system.

In future designs, we will also ensure that the nozzle is 11 ¼ " high, the most effective spray height according to our tests.

Trial #	Concentration of compost extract judged by color	Score
1		2
2		2
3		3
Average (Rounded)		2

Table 7: Results of Spray Nozzle at height of 15"

Trial #	Concentration of compost extract judged by color	Score
1		2
2		2
3		3
Average (Rounded)		2

Table 8: Results of Spray Nozzle at height of 13 5/8"

Trial #	Concentration of compost extract judged by color	Score
1		3
2		4
3		3
Average (Rounded)		3

Table 9: Results of Spray Nozzle at height of 11 1/2”

References

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Teissere, Ty. *Client Meeting Number 1*. Interview by Stella Ruebel et al., 4 Mar. 2026.

Teissere, Ty. *Client Meeting Number 2*. Interview by Victor Nyugen et al., 17 Apr. 2026.

Generative AI Statement

No generative AI was used on this assignment.